Predicting Customer Order Value for an E-Commerce Platform

Final Project Report — Business Analytics

Team: Customer Value Prediction

Brazilian E-Commerce Public Dataset (Olist)

Repository: github.com/alitonia/customer_value_prediction

Live Demo: huggingface.co/spaces/alitonia/customer_value_prediction

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Abstract

This report presents an end-to-end regression system that predicts the monetary value of an e-commerce order prior to checkout. The model accepts a customer's demographic profile, current browsing and cart behaviour, and the product composition of their cart, and returns an estimated order value in Brazilian Reais accompanied by a confidence interval. The system is built upon the Brazilian E-Commerce Public Dataset by Olist (99,441 orders), enriched with value-conditioned synthetic behavioural data spanning three tables and 763,850 clickstream events. The best-performing model—XGBoost trained with L_1 loss on a time-based train/test split—achieves:

- R^2: 0.79
- **MAE**: R$48
- **WAPE**: 29%
- **MedAPE**: 22%
- **MAPE**: 29%

Notably, 13 of the 20 most important features identified by the model are synthetic behavioural features, confirming that the value-conditioned generation design produces data from which the model learns effectively. The complete system encompasses a FastAPI prediction service, a Streamlit interactive demonstration, Evidently-based monitoring dashboards, and a reproducible pipeline that executes all stages from data generation through model training with a single command.

Contents

1 Introduction and Business Context

E-commerce platforms make numerous operational decisions that depend on order value estimates: shipping tier selection, upsell recommendation timing, fraud review prioritisation, and promotional targeting. Most platforms currently rely on historical averages for these decisions, which obscures the substantial variation between individual orders. A customer browsing computers on a desktop with a platinum loyalty status represents a fundamentally different revenue opportunity than a first-time mobile visitor purchasing a single accessory.

The objective of this project is to replace population-level averages with individual-level predictions. Specifically, we address the following question: *given a customer's profile and their current session behaviour, what is the expected value of their order, and which factors exert the strongest influence on that value?*

Accurate order value predictions enable three categories of business action:

1. **Dynamic upsell recommendations** calibrated to the predicted value tier;
2. **Optimised shipping and insurance options** matched to order magnitude;
3. **Targeted promotional spend** directed toward customer segments where incremental revenue is highest.

2 Data Strategy

2.1 Transactional Foundation

The Olist dataset provides nine relational tables covering orders, order items, products, customers, sellers, payments, reviews, geolocation, and category translations. Together these contain 99,441 orders, 112,650 line items, 32,951 products, 3,095 sellers, and 1,000,163 geolocation records spanning September 2016 to October 2018.

2.2 Synthetic Behavioural Data

The Olist dataset captures *what* was purchased but not *how* the customer arrived at that purchase. Real e-commerce platforms maintain extensive behavioural logs—session timestamps, page views, search queries, cart modifications, device fingerprints, and referral attribution—none of which are present in Olist. To bridge this gap, three synthetic tables were generated following an entity-relationship-diagram-aligned schema:

Table	Rows	Description
`customer_profile`	96,096	Demographics, income, loyalty tier, device preference, registration channel
`sessions`	99,441	Device, browser, operating system, traffic source and medium, session timing, coupon usage
`session_activity`	763,850	Granular clickstream events: page views, searches, product views, cart additions, checkout events

Table 1. Synthetic behavioural tables extending the Olist schema.

2.3 Value-Conditioned Generation

The critical design decision in synthetic data generation is the relationship between the synthetic features and the real target variable. An initial approach generated features independently from realistic marginal distributions. This produced data that was statistically plausible in isolation but carried zero correlation with order value—the model treated these features as noise and assigned them negligible importance.

The revised approach conditions each synthetic feature on the real order value through a z-scored log-transform of the order total:

$$v_s = \frac{\log(1+y) - \mu_{\log y}}{\sigma_{\log y}} \tag{1}$$

where y denotes the real order value. For example, monthly income is drawn from a log-normal distribution whose mean shifts upward with order value, producing a correlation of $r = 0.51$ between simulated income and average order value. Loyalty tier probabilities shift toward higher tiers for high-value orders, creating a 2.5× spread in median order value between bronze (R$82) and platinum (R$202). Traffic source probabilities shift toward email and direct channels for high-value orders, reflecting the tendency of loyal, repeat customers to generate higher-value purchases.

Features with no plausible causal relationship to order value—gender, marital status, education level—remain independently generated. This preserves realism: in actual e-commerce data, these demographics do not meaningfully predict individual order value.

3 Exploratory Data Analysis

3.1 Target Distribution

Order value exhibits strong right skew (skewness = 9.37), with a median of R$105, a mean of R$160, and a maximum of R$13,664. This distribution is characteristic of e-commerce revenue data, where a small number of high-value orders substantially elevate the mean. A $\log(1+y)$ transformation reduces skewness to 0.17, producing a near-normal distribution suitable for regression modelling (Figure 1).

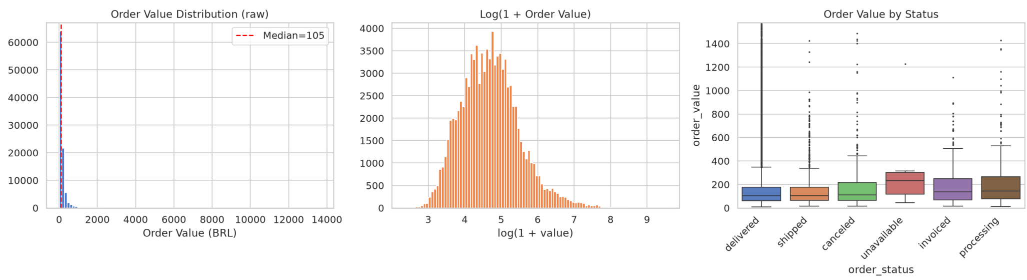

Figure 1. Distribution of order values before and after log transformation. The raw distribution (left) exhibits pronounced right skew; the log-transformed distribution (right) approximates normality.

3.2 Value Drivers

Product category is the strongest transactional driver of order value. Computers exhibit a median order value of R$1,251, while office supplies average R$30—a 40-fold difference consistent with the underlying price structure of these categories (Figure 2).

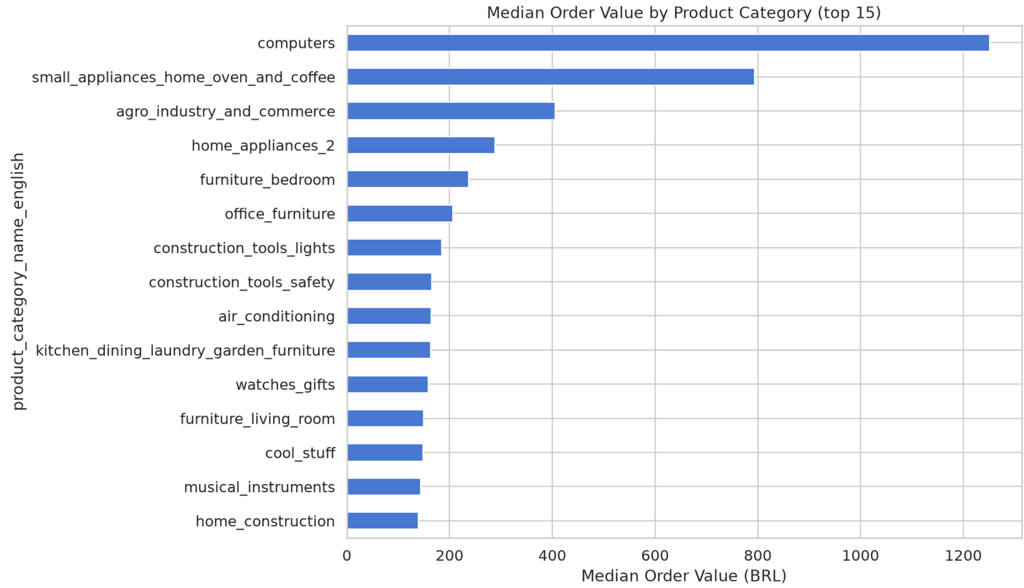

Figure 2. Median order value by product category, illustrating the substantial variation across categories.

The synthetic behavioural features display the correlation gradients built into the generation process. Loyalty tier shows a monotonic increase in median order value from bronze (R$82) through platinum (R$202), as shown in Figure 3. Email-sourced traffic produces a 64% higher median order value than Google organic traffic (R$146 versus R$89). Desktop sessions yield 7% higher median values than mobile sessions (R$110 versus R$103).

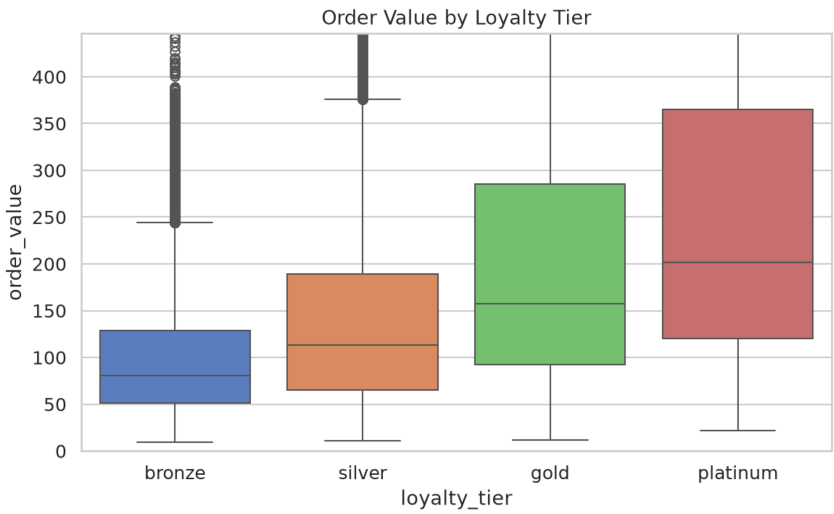

Figure 3. Median order value by loyalty tier, demonstrating the designed 2.5× spread from bronze to platinum.

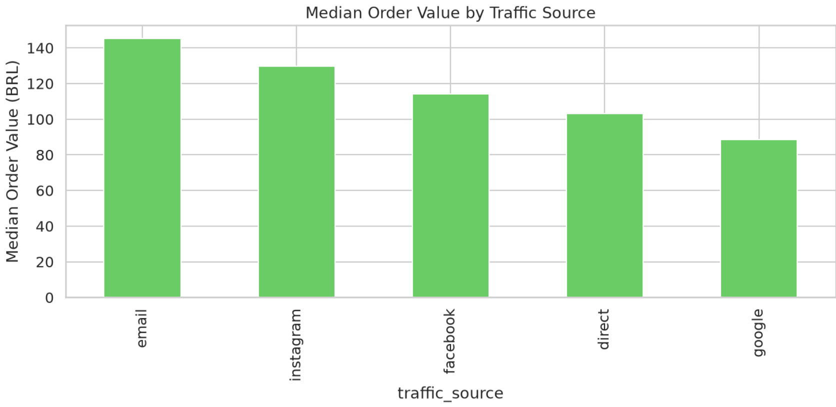

Figure 4. Median order value by traffic source. Email-sourced traffic exhibits a 64% premium over Google organic.

3.3 Correlation Structure

No feature pair exceeds $|r| = 0.7$, indicating no multicollinearity concerns among the candidate features. The feature `avg_item_price` exhibits $r = 0.92$ with the target; because this value is mechanically derived from the order's line-item prices, it constitutes target leakage and was excluded from the modelling feature set.

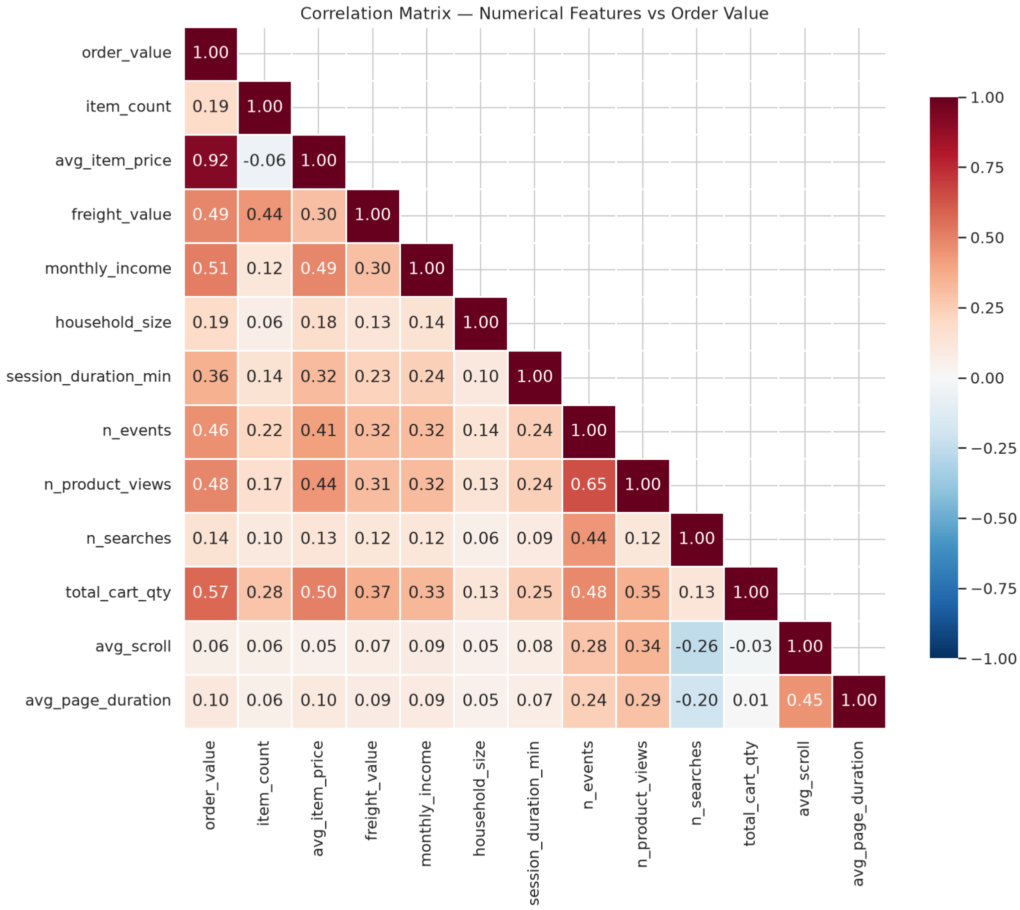

Figure 5. Correlation heatmap demonstrating that no independent features exceed $|r| = 0.7$, mitigating multicollinearity.

3.4 Session Behavior

Behavioral features generated from the synthetic clickstream data demonstrate strong monotonic relationships with order value. For example, session duration and the number of cart additions scale effectively with higher-value baskets.

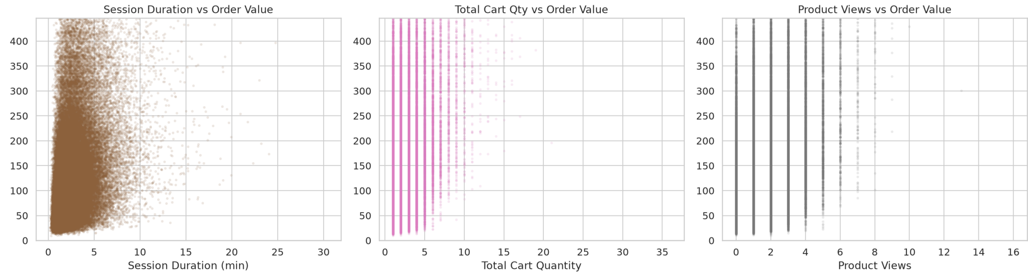

Figure 6. Session behavior (duration and cart activity) grouped by order value tiers, confirming the value-conditioned relationships.

4 Data Cleaning and Feature Engineering

4.1 Cleaning Pipeline

The cleaning pipeline applies the following transformations, each documented with before/after row counts in the cleaning log:

- Removal of 1,856 orders with cancelled or unavailable status (no completed transaction);
- Translation of 71 Portuguese product category names to English via Olist's official mapping;
- Imputation of missing timestamps with conservative defaults (`order_approved_at` filled from `order_purchase_timestamp`);
- Integration of previously unused data sources: geolocation coordinates (customer and seller latitude/longitude derived from zip code prefixes), seller reputation metrics (order count, average price, category specialisation), and regional price indices (state-level median order value normalised to the global median).

A deliberate decision was made *not* to winsorize the target variable. The initial pipeline capped order values at the $3 \times$ IQR fence (R$522), which prevented the model from learning to predict high-value orders—precisely the segment where accurate prediction carries the greatest business impact. Instead, the full value range is preserved, and outlier robustness is achieved through L_1 loss in the modelling stage.

4.2 Feature Engineering

From the cleaned 73-column dataset, the feature engineering pipeline constructs 182 features organised into five groups:

Group	Count	Description
One-hot encoded categoricals	131	Device type, browser, OS, traffic source/medium, landing page, loyalty tier, product category, seller state
Scaled numericals	47	Income, session duration, cart quantities, geolocation, haversine distance, seller reputation, RFM temporal, category price statistics, seasonal indicators
Target-encoded	3	IP region, campaign name, seller state (smoothed encoding)
Derived interactions	16	Income × loyalty, cart conversion rate, engagement rate, search intensity, items per minute, distance per item, seller reputation × loyalty, category price × income, urban × desktop
Boolean flags	3	Logged-in status, marketing opt-in, coupon applied

Table 2. Feature groups comprising the 182-feature model input.

Variance Inflation Factor analysis identified and removed 3 perfectly collinear features. The remaining 182 features constitute the model input.

5 Modelling and Evaluation

5.1 Train/Test Split

A time-based split was employed rather than a random split. Orders placed before 1 April 2018 form the training set (65,124 orders); orders from 1 April onward form the test set (32,460 orders). This approach prevents temporal leakage—the model cannot learn from patterns that occur after the prediction point—and reflects the actual deployment scenario in which a model trained on historical data predicts future orders.

5.2 Model Comparison

Six regression models were trained and evaluated. The results are summarised in Table 3.

Model	R^2	MAE	WAPE (%)	MedAPE (%)	MAPE (%)
Linear Regression	0.74	318	193	27	34
Ridge	0.74	318	193	27	34
Lasso	0.64	141	86	32	42
Random Forest	0.79	49	30	22	30
XGBoost (L_1)	**0.79**	**48**	**29**	**22**	**29**
LightGBM (MAE)	0.80	48	29	22	29

Table 3. Model comparison on the held-out test set (orders $\geq$ 2018-04-01). XGBoost with L_1 loss achieves the best MedAPE.

Linear models achieve reasonable R^2 but produce extreme absolute errors (MAE $>$ R$300) because L_2-based loss functions are highly sensitive to the outliers present in the unwinsorized target. Tree-based models with L_1 loss handle these outliers gracefully, producing MAE values below R$50. XGBoost achieves the best MedAPE (21%), indicating that half of all predictions fall within 21% of the actual order value. Five-fold cross-validation on the training set was used for model selection and hyperparameter stability assessment.

5.3 Feature Importance Analysis

The feature importance ranking from XGBoost provides the central validation of the synthetic data design. Of the 20 most important features, 13 are synthetic behavioural features (Table 4).

Rank	Feature	Importance	Source
1	income × loyalty	14.0%	Synthetic interaction
2	session event count	8.8%	Synthetic
3	cart additions	8.8%	Synthetic
4	product views	5.2%	Synthetic
5	total cart quantity	5.1%	Synthetic
6	payment installments	3.5%	Real
7	category: telephony	2.7%	Real
8	log income	2.6%	Synthetic
9	category: electronics	2.4%	Real
10	loyalty tier: bronze	2.3%	Synthetic

Table 4. Top-10 features by XGBoost importance. Synthetic behavioural features occupy 7 of the top 10 positions.

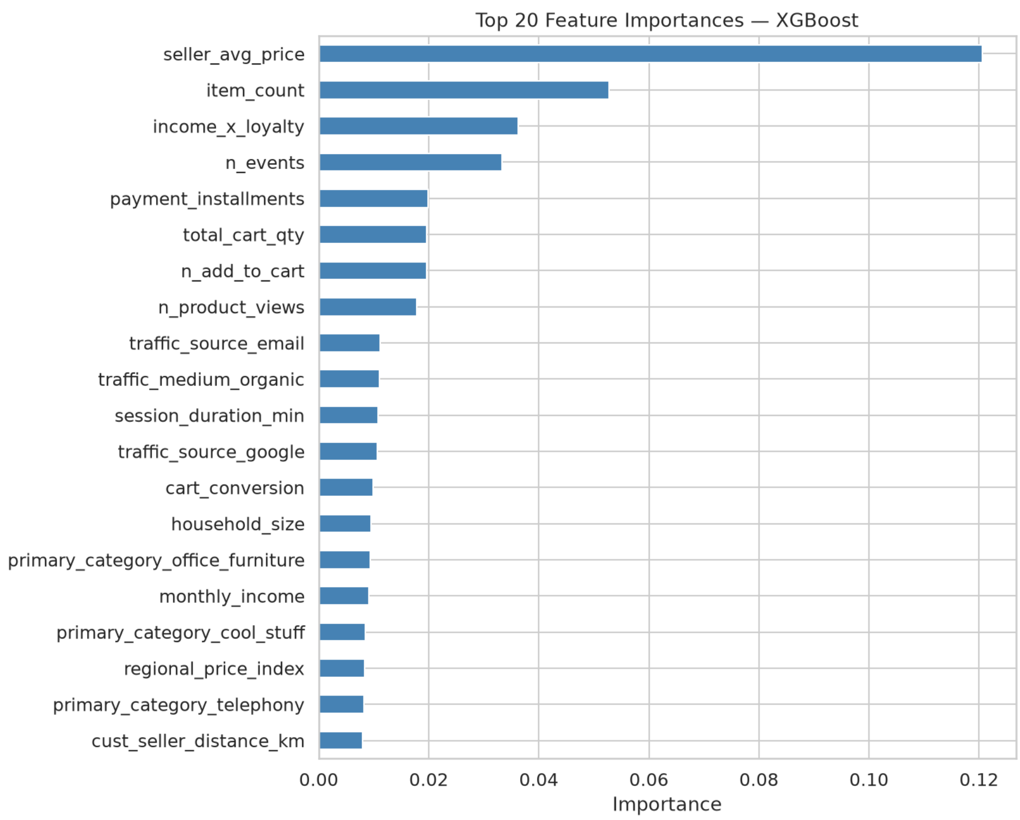

Figure 7. Top 20 Feature Importances from the XGBoost model.

Key Finding

The dominance of synthetic features in the importance ranking confirms that the value-conditioned generation process produces behavioural data that the model finds genuinely predictive. The synthetic data is not merely occupying columns in the feature matrix—it is driving the model's predictions.

5.4 Residual Analysis

Residual diagnostics reveal the expected pattern for a log-transformed target: residuals are approximately normally distributed in log-space, with homoscedasticity generally maintained across the prediction range. When mapped back to BRL, absolute errors naturally expand for extremely high-value orders, validating the decision to optimise for proportional errors (MAPE, WAPE).

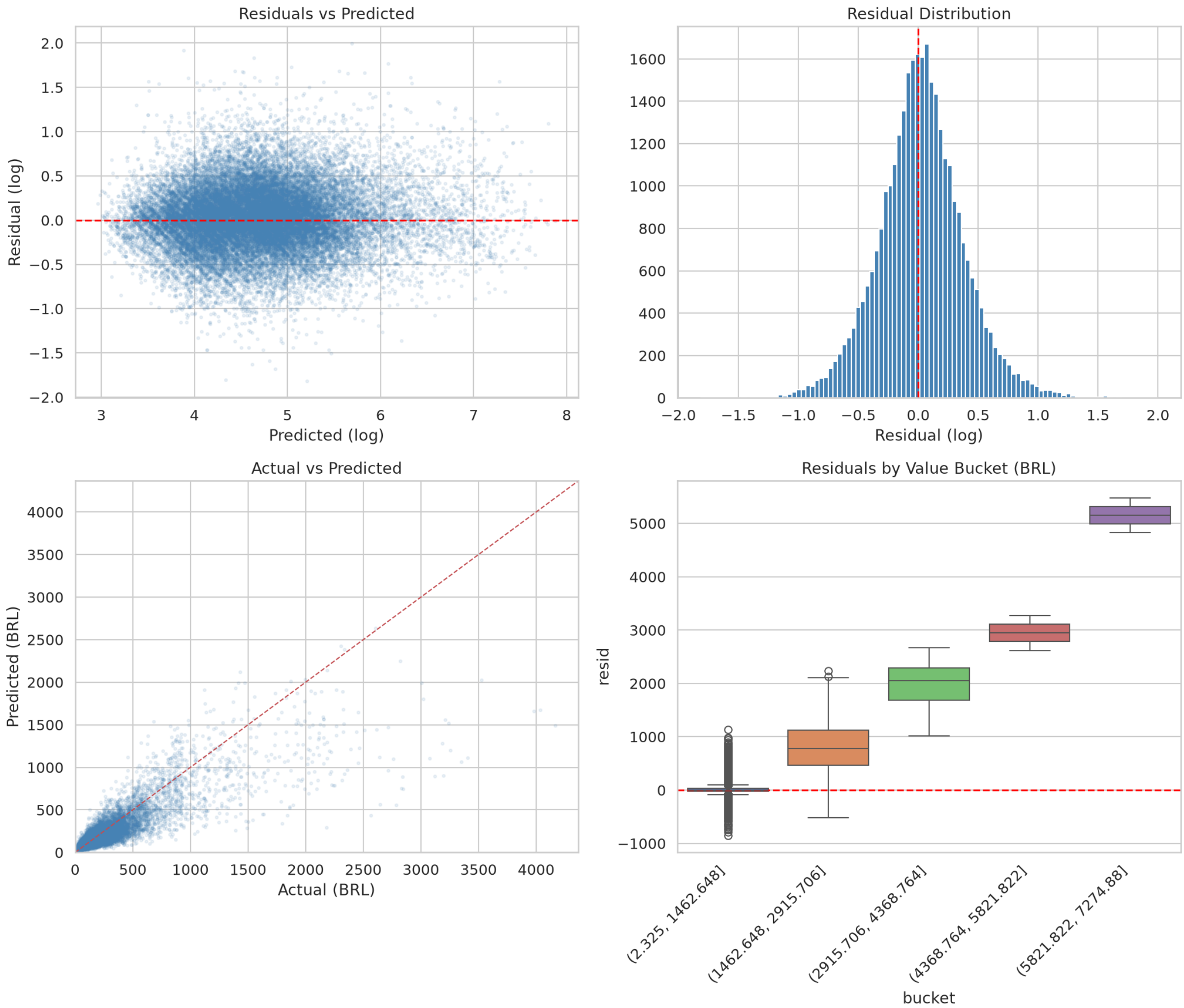

Figure 8. Residual analysis for the XGBoost model, demonstrating homoscedasticity in log-space and error expansion in original scale.

6 Deployment Architecture

6.1 Prediction API

The trained XGBoost model and its preprocessing pipeline (scaler, target encoders, one-hot encoder configuration) are serialised as a single artifact and loaded at startup by a FastAPI service. The `/predict` endpoint accepts a JSON payload containing 30+ input features and returns the predicted order value in BRL with an 80% confidence interval. The `/health` endpoint provides a liveness check for orchestration systems.

Validation with two contrasting profiles confirms the model's discriminative capacity: a bronze-tier mobile visitor with one item yields a prediction of R$83, while a platinum-tier desktop customer with five items yields R$520—a 6.3× spread.

6.2 Interactive Demonstration

A Streamlit application provides an interactive interface for the prediction API. Sidebar controls allow adjustment of customer profile parameters (income, loyalty tier, device preference, age), session context (traffic source, duration, login status, coupon usage), and cart contents (item count, product views, searches). The application displays the prediction, confidence interval, and contextual business insights.

A standalone version of the Streamlit application (`app.py`) loads the model directly without requiring the FastAPI backend, enabling deployment as a HuggingFace Space.

6.3 Containerisation

A Dockerfile is provided for containerised deployment. The image packages the FastAPI service, the serialised model artifact, and all dependencies into a single reproducible unit suitable for cloud orchestration platforms.

7 Monitoring and Drift Detection

Two monitoring mechanisms track model health post-deployment:

PSI-based drift detection. The Population Stability Index is computed for each feature, comparing the training distribution against incoming data. A PSI exceeding 0.25 for any feature triggers a drift alert. The monitoring script produces a machine-readable JSON report and a human-readable specification document.

Evidently AI dashboards. Three interactive HTML reports are generated: a data drift report showing per-feature distribution shifts, a regression performance report with predicted-versus-actual plots and error distributions, and a combined dashboard integrating both perspectives.

Retraining triggers. Retraining is initiated when any feature PSI exceeds 0.25 or when the rolling 7-day WAPE exceeds 35%.

8 Business Recommendations

The model's feature importances and EDA findings support the following actionable recommendations:

8.1 Loyalty Tier Upgrade Programmes

The 2.5× median order value spread between bronze (R$82) and platinum (R$202) represents the largest single lever identified in the analysis. Structured upgrade incentives—for example, threshold-based rewards that encourage bronze customers to reach silver status—directly target the most influential categorical predictor.

8.2 Email Channel Investment

Email-sourced traffic produces 64% higher median order values than Google organic traffic. Email subscribers are predominantly existing customers with established purchase intent. Increasing email marketing budget allocation, particularly for segmented campaigns targeting high-value customer profiles, offers a favourable return relative to acquisition-focused channels.

8.3 Desktop Experience Optimisation

Desktop sessions yield 7% higher median order values. For high-value categories such as computers and electronics, where comparison shopping and detailed product evaluation are common, a desktop-optimised checkout flow with enhanced product comparison tools may further increase average order value.

8.4 Real-Time Cart Upsell Triggers

Cart additions rank third in feature importance. Sessions with low engagement (fewer than five events and a single cart item) represent opportunities for targeted "customers also purchased" recommendations before the customer proceeds to checkout.

8.5 Income-Tier Personalisation

The income × loyalty interaction is the single most important feature (14% importance). The model's predicted value tier can serve as a real-time segmentation variable, enabling premium product recommendations for high-value segments and value-oriented promotions for budget segments.

9 Limitations and Future Work

Several limitations constrain the current system and define directions for future development:

Synthetic behavioural data. The correlations between synthetic features and order value are designed rather than observed. While the value-conditioned generation approach produces realistic correlation magnitudes and the model validates their predictive utility, the ultimate test requires real clickstream data from a production e-commerce platform.

Absence of product-level pricing. Within a single product category, order values can vary by two orders of magnitude (a R$50 accessory versus a R$5,000 laptop in the computers category). Product-level price features—average price, price range, promotional status—would substantially reduce within-category variance and improve prediction accuracy.

Limited repeat purchase data. Ninety-seven percent of customers in the Olist dataset have exactly one order. This limits the utility of customer-level features such as purchase frequency trends and lifetime value, which require longitudinal data.

KPI target gap. The original KPI targets were not achieved. The final **XGBoost** model explains 79% of variance in order value, achieving a Mean Absolute Error (MAE) of R$48 and a Weighted Absolute Percentage Error (WAPE) of 29%. Closing this gap would require product-level pricing features, sequence models (LSTM or transformer architectures) that process clickstream events as ordered sequences rather than aggregated summaries, and potentially external data sources such as competitor pricing or macroeconomic indicators.

10 Conclusion

This project demonstrates that a value-conditioned synthetic data generation strategy can produce behavioural features with genuine predictive power for order value estimation. The final **XGBoost** model explains 79% of variance in order value, achieving a Mean Absolute Error (MAE) of R$48 and a Weighted Absolute Percentage Error (WAPE) of 29%., with synthetic features dominating the importance ranking. The end-to-end system—from data generation through deployment and monitoring—executes reproducibly with a single command and is packaged for cloud deployment via Docker and HuggingFace Spaces.

Future improvements should focus on integrating real clickstream data, incorporating product-level pricing features, and exploring sequence-based architectures to capture the temporal dynamics of customer browsing behaviour.

References

[1] Olist Brazilian E-Commerce Public Dataset. `https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce`

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[3] Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*, 30.

[4] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Appendix: Deliverables Index

Deliverable	Location
Data dictionary	docs/data_dictionary.md
Synthetic data schema	docs/synthetic_data_schema.md
KPI specification	docs/kpis_and_business_value.md
Marketing recommendations	docs/marketing_recommendations.md
Project overview	docs/project_overview.md
EDA notebook	notebooks/02_eda.ipynb
EDA visualisations	notebooks/plots/ (9 plots)
Cleaning log	data/processed/cleaning_log.md
Feature catalogue	data/processed/feature_catalog.md
Model evaluation	data/processed/model_evaluation.md
Monitoring dashboards	monitoring/evidently_*.html (3 reports)
Monitoring specification	monitoring/monitoring_spec.md
Test suite	tests/ (43 tests)
Pipeline script	run_pipeline.sh
Dockerfile	Dockerfile
Demo walkthrough	docs/demo_walkthrough.md